

Data-Driven Discovery of Power Grid Frequency Dynamics using Symbolic Regression

FELIX WANG*, Karlsruhe Institute of Technology (KIT), Germany

MAILY DOAN*, Karlsruhe Institute of Technology (KIT), Germany

XINYI WEN, Karlsruhe Institute of Technology (KIT), Germany

BENJAMIN SCHÄFER, Karlsruhe Institute of Technology (KIT), Germany

The transition towards renewable energy sources and the resulting decentralization of power generation pose significant challenges to grid stability, particularly due to the reduction of system inertia. While traditional physics-based models provide a robust theoretical framework, they often rely on rigid assumptions that may not fully capture the complex, weather-dependent fluctuations of modern power grids. This paper investigates the potential of Symbolic Regression (SR) as a purely data-driven approach to discover the governing differential equations of frequency dynamics directly from measurement data, without presupposing specific physical laws. We systematically evaluate and compare four distinct SR algorithms (SINDy, PySR, AI-Feynman, DSR), assessing their capability to reconstruct known physical dynamics from synthetic datasets and to identify governing equations from empirical South Korean grid frequency data. To address the inherent stochasticity of power systems, we propose a hybrid modeling approach: After extracting the deterministic core dynamics using SR on Gaussian filtered data, we analyze the model residuals to fit specific noise distributions, to capture heavy-tailed statistics observed in real-world grids. Our comparative analysis demonstrates that most algorithms efficiently recover sparse, interpretable models. The methods identify non-linear terms often overlooked by classical linear models. Furthermore, the proposed hybrid model accurately reproduces key statistical properties, including probability density functions and autocorrelation patterns, offering a more realistic tool for monitoring stability in low-inertia power systems.

CCS Concepts: • **Applied computing** → **Engineering; Mathematics and statistics; Engineering**; • **Hardware** → **Power and energy; Power estimation and optimization; Power networks**.

ACM Reference Format:

Felix Wang, Maily Doan, Xinyi Wen, and Benjamin Schäfer. 2026. Data-Driven Discovery of Power Grid Frequency Dynamics using Symbolic Regression. 1, 1 (March 2026), 7 pages. <https://doi.org/10.1145/nnnnnnn.nnnnnnn>

*Both authors contributed equally to this research.

Authors' Contact Information: Felix Wang, felix-wang@outlook.de, Institute for Automation and Applied Informatics (IAI), Karlsruhe Institute of Technology (KIT), Eggenstein-Leopoldshafen, Germany; Maily Doan, mailydoan@web.de, Institute for Automation and Applied Informatics (IAI), Karlsruhe Institute of Technology (KIT), Eggenstein-Leopoldshafen, Germany; Xinyi Wen, xinyi.wen@kit.edu, Institute for Automation and Applied Informatics (IAI), Karlsruhe Institute of Technology (KIT), Eggenstein-Leopoldshafen, Germany; Benjamin Schäfer, benjamin.schaefer@kit.edu, Institute for Automation and Applied Informatics (IAI), Karlsruhe Institute of Technology (KIT), Eggenstein-Leopoldshafen, Germany.

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than the author(s) must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from permissions@acm.org.

© 2026 Copyright held by the owner/author(s). Publication rights licensed to ACM.

ACM XXXX-XXXX/2026/3-ART

<https://doi.org/10.1145/nnnnnnn.nnnnnnn>

1 Introduction

The ongoing transition toward renewable energy sources is fundamentally reshaping the operation of electric power systems. While the replacement of fossil-fuel-based generation with wind and solar power is essential to mitigate climate change, these sources introduce weather-dependent volatility and uncertainty into the power supply [Staffell and Pfenninger 2018]. At the same time, the displacement of conventional synchronous generators reduces the rotational inertia available in the system, making the grid more susceptible to frequency deviations [Ulbig et al. 2015]. The power grid frequency serves as the central indicator of the instantaneous balance between generation and load. A surplus of generation causes the frequency to rise above its nominal value, while a deficit leads to a drop. When these deviations grow too large, they increase the risk of cascading failures and, in extreme cases, large-scale blackouts [Anvari et al. 2016; Ulbig et al. 2015]. Consequently, as renewable penetration grows and system inertia declines, a deeper understanding of frequency dynamics becomes increasingly important for developing robust monitoring and control strategies [Homan et al. 2021].

The dynamics of the power grid frequency are shaped by the interplay of deterministic processes, such as generation scheduling and control actions, with stochastic perturbations arising from load fluctuations, forecast errors, and volatile renewable feed-in [Gorjao et al. 2020]. Classical approaches to modeling these dynamics rely on physics-based formulations rooted in the aggregated swing equation, which describes the frequency response in terms of system inertia, primary and secondary control, and power imbalance [Gorjao et al. 2020]. While this framework has proven highly useful, its application to modern power systems faces growing challenges. The increasing complexity and decentralization of the grid make detailed bottom-up models difficult to construct, as access to the parameters and data of all connected devices is rarely available [citation 14 from Wen et al. 2024; citation 23 from Wen et al. 2024]. Moreover, effective frequency dynamics at the system level may exhibit nonlinear behavior that arises from the interplay of individually linear control mechanisms [Oberhofer et al. 2023] [+ citation 36 from Wen et al. 2024], rendering predefined model structures potentially incomplete.

In response to these challenges, data-driven modeling has gained increasing attention as a complementary paradigm. As discussed by Huang et al. [2023], data-driven approaches for power system dynamics can be organized along a spectrum of reliance on prior physical knowledge. At one end, physics-inspired model construction creates model structures guided by physical insight and experience. At the other end, purely data-driven methods synthesize models directly from measurement data with little to no prior structural assumptions. Between these extremes, hybrid approaches integrate

physical knowledge with data-driven techniques. Recent stochastic frequency models illustrate this progression. Kraljic [2023] proposed a phenomenological model for the Great Britain grid that modifies the swing equation based on observed statistical patterns, incorporating Fractional Lévy Noise to reproduce the heavy-tailed distributions and long-range correlations found in empirical frequency data. Kruse et al. [2022] developed a hybrid techno-economic model for the Continental European grid, in which a neural network maps external features such as system load and renewable generation share to time-varying parameters of a stochastic differential equation. Both approaches successfully reproduce important statistical properties of the grid frequency, yet they rely on predefined model structures that must be manually specified and adapted.

Symbolic regression (SR) offers a fundamentally different perspective. Rather than fitting parameters within a prescribed model structure, SR algorithms search the space of mathematical expressions to discover governing equations directly from data [Brunton et al. 2016; Cranmer 2023]. This makes SR a purely data-driven approach to model construction in the taxonomy of Huang et al. [2023], with the distinctive advantage of producing interpretable, closed-form equations rather than black-box predictions. In the power systems domain, SR methods have been applied to various problems, including the identification of nonlinear dynamics in synchronous generators [Saria et al. 2021; Stankovic et al. 2020], the detection of nonlinear energy flows in grids with high inverter-based renewable penetration [Saeed Kandezy et al. 2024], model predictive control for voltage stability [Tran et al. 2023], and voltage profile prediction using physics-informed operators [Eichhorn et al. 2025]. For power grid frequency dynamics specifically, Wen et al. [2024] recently applied the Sparse Identification of Nonlinear Dynamics (SINDy) method to empirical frequency data from Iceland, Ireland, and the Balearic Islands. Their study demonstrated that SINDy can recover the expected primary and secondary control terms from measurement data and, notably, identified nonlinear terms such as ω^2 and $\theta\omega$ that are not present in standard linear models. This work established SR as a viable tool for frequency dynamics discovery.

Different SR paradigms embody fundamentally different inductive biases and search strategies, which may lead to different insights when applied to the same physical system. Sparse regression methods such as SINDy assume that the dynamics are a sparse linear combination of terms from a predefined function library and identify relevant terms through regularization [Brunton et al. 2016]. Genetic programming approaches such as PySR evolve populations of candidate expressions through mutation and crossover, searching an open-ended space of mathematical forms without requiring a fixed library [Cranmer 2023]. Bayesian variational inference methods such as SVISE jointly estimate the latent system state and the governing equations from noisy observations, handling measurement noise within a probabilistic framework rather than requiring explicit pre-filtering [Course and Nair 2023]. Whether these paradigmatic differences matter in practice for power grid frequency modeling remains an open question.

In this paper, we present the first systematic comparison of three distinct SR paradigms for discovering the governing equations of power grid frequency dynamics. We apply SINDy, PySR, and SVISE

to both synthetic data with known ground-truth dynamics and empirical frequency recordings from the South Korean power grid. The synthetic validation allows us to assess each method's ability to recover known equation structures and coefficients under controlled conditions, including varying noise levels. The application to empirical data from a previously unstudied grid provides new insights into the frequency dynamics of the South Korean system and tests whether the different SR methods agree on the discovered governing equations. To the best of our knowledge, this is also the first application of SR methods to frequency data from an Asian mainland grid.

The remainder of this paper is organized as follows. Section 2 reviews related work on stochastic frequency models and symbolic regression in power systems. Section 3 introduces the stochastic model of frequency dynamics that serves as the theoretical reference and validation basis. Section 4 describes the three SR methods employed. Section 5 details the experimental setup, including data, preprocessing, and evaluation metrics. Section 6 presents the results on both synthetic and empirical data. Section 7 discusses the findings, and Section 8 concludes.

2 Related Work

2.1 Stochastic Models of Power Grid Frequency

2.2 Symbolic Regression in Power Systems

3 Modeling Framework

The frequency dynamics of a synchronous power grid can be described through an aggregated swing equation that captures the collective behavior of all generators and loads connected to the system. Following Gorjao et al. [2020], we express the dynamics in the rotating reference frame using the bulk angular velocity $\omega = 2\pi(f - f_{\text{ref}})$, which measures the deviation of the grid frequency f from the nominal reference frequency f_{ref} (50 Hz or 60 Hz depending on the grid). The bulk angle θ , defined as the time integral of ω , represents the accumulated phase deviation. In this formulation, the system dynamics are governed by the coupled stochastic differential equations

$$\frac{d\theta}{dt} = \omega, \quad (1)$$

$$M \frac{d\omega}{dt} = -c_1\omega - c_2\theta + \Delta P + \epsilon\xi, \quad (2)$$

where M denotes the inertial constant of the system, which sets the time scale of the frequency response. For simplicity, M can be absorbed into the remaining parameters by dividing Equation (2) throughout, yielding rescaled coefficients c_1/M , c_2/M , $\Delta P/M$, and ϵ/M [Gorjao et al. 2020]. In the following, we adopt this normalization and set $M = 1$.

The individual terms in Equation (2) represent distinct physical mechanisms. The term $-c_1\omega$ models primary (proportional) control and general damping; the larger the frequency deviation ω , the stronger the restoring force that drives the system back toward its nominal frequency [Gorjao et al. 2020]. The term $-c_2\theta$ represents secondary (integral) control, which acts on the accumulated phase deviation to ensure long-term restoration of the nominal frequency after sustained disturbances. Because primary control responds on

the time scale of seconds while secondary control operates over minutes, the parameters typically satisfy $c_2 \ll c_1$ [Gorjao et al. 2020]; citation 50 from Gorjão et al. 2020]. The deterministic power mismatch ΔP captures the imbalance between scheduled generation and actual load demand. In practice, ΔP exhibits characteristic jumps at regular dispatch intervals (typically every 15 to 60 minutes), when the step-wise generation schedule is updated to track the continuously evolving load curve [citation 24 from Gorjão et al. 2020; Kruse et al., 2022]. Within each dispatch interval, ΔP can be approximated as a piecewise function that ramps between the scheduled steps. Finally, $\epsilon \xi$ represents the aggregated stochastic perturbation acting on the system, where ξ is typically modeled as Gaussian white noise and ϵ controls the noise amplitude. This term subsumes diverse fluctuation sources, including demand variability, intermittent renewable generation, and forecast errors [Gorjao et al. 2020].

This general model serves as the foundation for more specialized formulations found in the literature. Kraljic [2023] replaces the Gaussian noise with Fractional Lévy Noise to capture heavy-tailed distributions and long-range correlations, while absorbing secondary control effects into an empirically determined restoring function. Kruse et al. [2022] retain the linear structure of Equations (1) and (2) but allow the parameters c_1 , c_2 , ΔP , and ϵ to vary over time, with their values predicted by a neural network from techno-economic features. Both approaches demonstrate that the swing equation framework is flexible enough to accommodate diverse grid characteristics when suitably parameterized.

In our study, this model fulfills two roles. First, it provides the ground truth for generating synthetic validation data. Following the methodology of Wen et al. [2024], we employ the one-dimensional linear Kramers–Moyal model (1D-L-KM) introduced by Oberhofer et al. [2023], which is a specific instantiation of Equations (1)–(2) with linear control terms and a piecewise constant power mismatch $\Delta P = P_0$. Synthetic time series are generated using the Euler–Maruyama algorithm, with ground-truth values for c_1 and c_2 obtained through estimation of the Kramers–Moyal coefficients from empirical data [Gorjão and Meirinhos 2019]. By generating datasets both without noise ($\epsilon = 0$) and with Gaussian noise ($\epsilon > 0$), we can assess the ability of each SR method to recover the known equation structure and coefficients under controlled conditions. Second, the model serves as the interpretive reference against which the equations discovered by the SR algorithms on empirical data are evaluated. The physically motivated terms $-c_1\omega$ and $-c_2\theta$ provide a baseline expectation for the equation structure. Any additional terms identified by the SR methods, such as the nonlinear contributions ω^2 and $\theta\omega$ reported by Wen et al. [2024], can then be assessed relative to this established framework.

4 Symbolic Regression Methods

We employ three SR methods, each representing fundamentally different approaches to identifying governing equations from data. This diversity enables us to examine whether the choice of SR paradigm influences the dynamics discovered by the models. In the following subsections, we present the three methods applied in this study.

4.1 Sparse Identification of Nonlinear Dynamics (SINDy)

4.2 PySR

4.3 Stochastic Variational Inference for State Estimation (SVISE)

Both SINDy and PySR require a numerical approximation of the time derivative $\dot{\omega}$ as their regression target. Because ω is recorded as a noisy discrete signal at 1 Hz, a Gaussian filter is applied to smooth the measurements before the derivative is computed (see Section 5.2). This preprocessing step introduces a methodological dependency, as the choice of filter kernel width σ determines how aggressively noise is suppressed and, consequently, how much fast transient behavior is retained in the signal. SVISE avoids this dependency entirely by treating the governing equations and the underlying state trajectory as jointly unknown quantities to be inferred from the raw noisy observations [Course and Nair 2023]. Rather than requiring a pre-computed derivative target, SVISE reconstructs a smooth state trajectory and identifies the governing equation simultaneously within a unified Bayesian framework. Originally developed as a method for latent path inference in nonlinear SDEs [Course and Nair 2023], SVISE models the temporal evolution of the state vector $\mathbf{x}_t = (\theta_t, \omega_t)^\top$ as a stochastic differential equation of the form

$$d\mathbf{x}_t = f_\theta(\mathbf{x}_t, t) dt + \Sigma_\theta d\boldsymbol{\beta}_t, \quad (3)$$

where f_θ denotes the unknown drift function governing the deterministic dynamics, Σ_θ is the diffusion matrix, and $\boldsymbol{\beta}_t$ represents a standard Brownian motion process. The drift function is parametrized as a sparse linear combination of predefined basis functions,

$$f_\theta(\mathbf{x}_t, t) = (\boldsymbol{\psi}(\mathbf{x}_t) \otimes \mathbf{I}) \boldsymbol{\theta}, \quad (4)$$

where $\boldsymbol{\psi}(\mathbf{x}_t)$ is a row vector of candidate basis functions evaluated at the current state and $\boldsymbol{\theta}$ collects the corresponding coefficients. In this regard, the basis function library in SVISE serves a similar role to the candidate library in SINDy, and both methods share the assumption that the true dynamics are sparse in the chosen basis. The key difference lies in how the coefficients are estimated and how uncertainty is handled.

Given a time-series dataset $\mathcal{D} = \{(t_i, \mathbf{y}_{t_i})\}_{i=1}^N$ of noisy observations, SVISE performs approximate Bayesian inference, specifically stochastic variational inference (VI), over both the latent state trajectory $\mathbf{x}_t$ and the governing-equation parameters $\boldsymbol{\theta}$ simultaneously. The posterior process over the state is approximated by a Markov Gaussian process whose marginal statistics, the mean $\mathbf{m}_\phi(t)$ and covariance $\mathbf{S}_\phi(t)$, are directly parametrized using radial basis function models. The parameters $\boldsymbol{\theta}$ of the drift function are assigned a sparsity-promoting half-Cauchy prior, encouraging parsimonious equation structures [Course and Nair 2023].

The variational parameters ϕ are optimized by maximizing the evidence lower bound (ELBO), which balances fidelity to the observed data against consistency with the prior dynamics. A central contribution of SVISE is a reparametrization trick for Markov Gaussian processes that expresses the ELBO entirely in terms of the directly parametrized marginal statistics $\mathbf{m}_\phi$ and $\mathbf{S}_\phi$, rather than requiring forward integration of the underlying SDE. This reformulation replaces the computationally expensive ODE solves that would otherwise be needed at each optimization iteration with stochastic

Table 1. Comparison of the three symbolic regression methods along key methodological dimensions.

	SINDy	PySR	SVISE
Paradigm	Sparse regr.	Genetic progr.	Bayesian VI
Search space	Fixed library	Open (operators)	Fixed library
Derivative req.	Yes	Yes	No (joint est.)
Noise handling	Pre-filtering	Pre-filtering	Built-in (prob.)
Output	Sparse coeffs.	Pareto front	Posterior + coeffs.

approximations that decompose into fully independent evaluations, making them well suited for parallel computation [Course and Nair 2023].

For the application to power grid frequency dynamics, this design has two practical consequences. First, because SVISE operates on the raw observations, it provides an independent analysis path that does not share the preprocessing assumptions of SINDy and PySR. Comparing the equations discovered through both paths serves as a consistency check: if SVISE and the filter-dependent methods converge on the same equation structure, confidence in the results is strengthened; if they diverge, the role of the Gaussian filter becomes an important point of discussion. Second, SVISE provides posterior distributions over the equation coefficients rather than point estimates, which allows the significance of individual terms to be assessed through the learned uncertainty. Course and Nair [2023] demonstrated that SVISE outperforms several SINDy variants in benchmark problems, particularly in regimes with high observation noise and limited data, conditions that are relevant to the analysis of empirical frequency recordings.

Table 1 summarizes the key characteristics of the three methods. SINDy and SVISE both operate on a predefined basis function library, which ensures that the discovered equations remain interpretable but restricts the search to functional forms that the user has anticipated. PySR, by contrast, constructs expressions from a set of elementary operators without a fixed library, allowing it to discover unexpected functional forms at the cost of a larger and less constrained search space. A particularly important distinction for the present study is the treatment of measurement noise. SINDy and PySR require Gaussian pre-filtering of the frequency signal before derivative estimation, whereas SVISE handles noise internally through its probabilistic formulation. This difference defines two independent preprocessing paths in our experimental setup (Section 5.2) and provides a natural basis for assessing the robustness of the discovered equations.

5 Data and Experimental Design

5.1 Data Overview

The empirical dataset consists frequency measurements from the South Korean power grid, recorded by an Electric Data Recorder (EDR) at the Korea Institute of Energy Technology (KEN-TECH). The measurement infrastructure and data collection follow the methodology described by [Jumar et al., 2021 from Oberhofer et al. 2025] and Rydin Gorjão et al. [2020]. The dataset covers the period from

August 15, 2024 to August 31, 2025 (382 days), sampled at 1 Hz without any temporal gaps, yielding 33 004 800 rows in total. Of these, 32 172 226 contain valid frequency readings (97.5 %), while the remaining 832 574 entries (2.5 %) are flagged by the quality indicator and excluded from our further analysis.

South Korea operates a 60 Hz synchronous grid with no AC interconnections to neighboring countries, which makes it functionally comparable to an island system despite its continental geography [Oberhofer et al. 2025]. The generation mix is dominated by coal, gas, and nuclear power, each contributing roughly 30–35 %, while renewable energy currently accounts for approximately five percent [Ishida, 2023 from Oberhofer et al. 2025].

Figure 1 shows the probability density of the measured frequency on both linear and logarithmic scales. The distribution is concentrated around a mean of $\mu = 60.003$ Hz with a standard deviation of $\sigma = 0.020$ Hz, and exhibits a visible bimodal structure with two distinct peaks. A slight negative skewness ($s = -0.219$) indicates that downward deviations from the nominal frequency occur more frequently than upward deviations of equal magnitude. The kurtosis of $\kappa = 2.326$ falls below the Gaussian reference value of 3, reflecting the flattened shape that arises from the bimodal structure of the distribution. These distributional properties align closely with the values reported by Oberhofer et al. [2025] for the South Korean grid ($s = -0.250 \pm 0.191$, $\kappa = 2.270 \pm 0.214$). The bimodal and non-Gaussian character of the distribution is consistent with the nonstandard frequency statistics observed across several Asian and Pacific grids by Wen et al. [2023]. As Oberhofer et al. [2025] demonstrate, a nonlinear (cubic) drift term is required to reproduce this bimodality in stochastic Fokker–Planck models, providing independent evidence that the underlying dynamics are inherently nonlinear.

The autocorrelation function (ACF), shown in Figure 2, displays a rapid initial decay within the first few minutes, followed by a slow, persistent decline that remains positive beyond 120 minutes ($\text{ACF} \approx 0.047$). This long-range temporal dependence is further confirmed by an estimated Hurst exponent of $H \approx 0.86$, obtained via detrended fluctuation analysis (DFA-1) on the full dataset. Oberhofer et al. [2025] report $H = 0.77 \pm 0.03$ for their South Korean grid dataset using the same method. Both values lie well above $H = 0.5$ and indicate pronounced positive correlations. Long-range correlations in grid frequency have previously been attributed to the interplay between stochastic power fluctuations and deterministic control mechanisms operating on multiple timescales [Kraljic 2023].

Together, the bimodal distribution, platykurtic shape, and persistent autocorrelation structure indicate that the South Korean grid frequency dynamics involve nonlinear processes that cannot be adequately captured by simple linear stochastic models. These statistical features provide a natural motivation for applying symbolic regression, which can discover governing equations from the data without presupposing a specific functional form.

5.1.1 Synthetic Data. To validate the SR methods under controlled conditions before applying them to empirical data, we generate synthetic frequency time series from the aggregated swing equation introduced in Section ?? . Following the one-dimensional linear

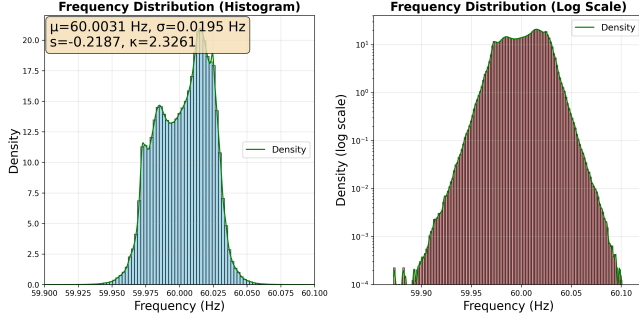


Fig. 1. Probability density of the South Korean grid frequency over the full recording period (August 2024 – August 2025). Left: linear scale, showing the bimodal structure and a kernel density estimate (green line). Right: logarithmic scale, revealing the tail behavior. Summary statistics are annotated in the left panel (μ : mean, σ : standard deviation, s : skewness, κ : Pearson kurtosis).

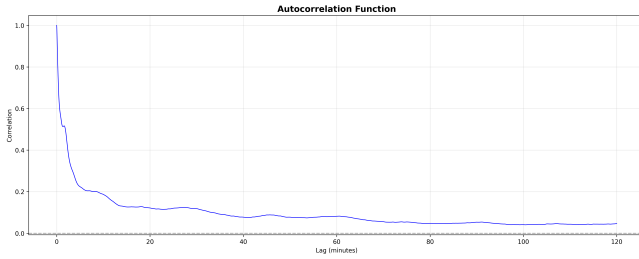


Fig. 2. Autocorrelation function of the South Korean grid frequency, computed on all valid data points ($N \approx 32.2 \times 10^6$) for lags up to 120 minutes. The rapid initial decay is followed by a persistent positive correlation, consistent with the Hurst exponent of $H \approx 0.86$.

Kramers–Moyal (1D-L-KM) model of ?, we simulate the dynamics

$$\frac{d\omega}{dt} = c_\omega \omega + \epsilon \xi(t), \quad (5)$$

where c_ω denotes the linear drift coefficient representing primary frequency control, ϵ is the noise amplitude, and $\xi(t)$ is Gaussian white noise. The parameters c_ω and ϵ are obtained from Kramers–Moyal coefficient estimation applied to the empirical South Korean data using the `kramersmoyal` package [Gorjão and Meirinhos 2019]. Time integration is performed with the Euler–Maruyama scheme at a step size of $\Delta t = 1$ s, matching the empirical sampling rate.

We consider two synthetic scenarios. In the *noise-free* case ($\epsilon = 0$), the time series is purely deterministic and follows $\dot{\omega} = c_\omega \omega$ exactly. This scenario tests whether each SR method can recover the correct equation structure without interference from stochastic fluctuations. In the *noisy* case ($\epsilon > 0$), we use the empirically estimated noise amplitude to produce time series that resemble realistic grid frequency recordings. This scenario assesses the robustness of each method to measurement and process noise, which is the primary practical challenge when applying SR to real-world grid data.

5.2 Preprocessing

The raw frequency recordings contain occasional measurement gaps and faulty values. We first apply a quality indicator filter provided with the dataset, which flags unreliable measurements. All flagged samples are removed prior to further processing. For short gaps of at most ten consecutive seconds, we apply time-based linear interpolation to maintain continuity. Longer gaps are left unimputed at this stage. During segmentation, only data chunks with a complete set of 300 samples (five minutes at 1 Hz) and no remaining missing values are retained for further analysis and training.

From the cleaned frequency signal $f(t)$, we compute the angular velocity deviation

$$\omega(t) = 2\pi(f(t) - f_{\text{ref}}), \quad (6)$$

where $f_{\text{ref}} = 60$ Hz is the nominal frequency of the South Korean grid. The bulk phase angle is obtained as the cumulative integral

$$\theta(t) = \int_0^t \omega(\tau) d\tau \approx \sum_{k=0}^t \omega(k) \Delta t \quad (7)$$

with $\Delta t = 1$ s at the given recording resolution. Together, θ and ω form the two-dimensional state vector $\mathbf{x} = [\theta, \omega]^\top$ that serves as input for all symbolic regression methods.

The empirical data is then partitioned into non-overlapping segments to account for the fact that grid dynamics vary over time and experience jumps in the mean frequency, caused by market dispatch events [Schafer et al. 2018]. In our study, we use segments of five minutes across models, since we found that these shorter windows improve training stability for SVISE.

PySR requires pre-smoothed input data, as its regression operates directly on the state variables and their numerical derivatives, both of which are sensitive to measurement noise. Following the approach of Wen et al. [2024], we apply a one-dimensional Gaussian filter to $\omega(t)$ prior to computing $\theta(t)$. The kernel width was tuned over $\sigma \in \{5, 10, 20, 60\}$, with $\sigma = 20$ yielding the best results. After smoothing, the time derivative is approximated via central finite differences,

$$\left. \frac{d\omega}{dt} \right|_{t_k} \approx \frac{\omega(t_{k+1}) - \omega(t_{k-1}))}{2\Delta t}, \quad (8)$$

providing the regression target for PySR, which learns the mapping $d\omega/dt = f(\theta, \omega, t)$. SVISE, by contrast, does not require external denoising or derivative estimation. Its variational inference framework treats measurement noise as an explicit model component, estimating state trajectories and their temporal derivatives jointly with the governing equations from noisy observations [Course and Nair 2023]. This distinction represents a key methodological difference and is reflected in the results.

5.3 Model Configuration

Each symbolic regression method involves a number of hyperparameters that influence the quality and stability of the discovered equations. We therefore conducted systematic hyperparameter tuning for each method on the empirical data. To keep the tuning computationally feasible, we restrict it to five-minute segments drawn from the 09:00 to 10:00 time window, which provides a representative

Table 2. Hyperparameter search space for SVISE and PySR. Best values are shown in **bold**.

Param.	Search values	Best	Description
<i>SVISE</i> (4,050 configurations, 60 sampled)			
σ	0, 5, 10	10	Smoothing width
d	2, 3	3	Library degree
τ	5e-1, 1e-1, 1e-2, 1e-3, 1e-5	5e-1	Sparsity prior scale
η	1e-1, 1e-2, 1e-3, 1e-4, 1e-5	1e-3	Learning rate
n_τ	50, 100, 200	100	Prior basis functions
σ_m	1e-3, 1e-4, 1e-5	1e-5	Meas. noise level
n_{rep}	15, 30, 50	30	Reparam. samples
<i>PySR</i> (3,888 configurations)			
σ	5, 10, 20, 60	20	Smoothing width
Tourn.	10, 20, 30	20	Selection pool size
Parsim.	1e-3, 1e-2, 1e-1, 0	0	Complexity penalty
Cycles	100, 300, 500	100	Cycles per iteration
Iters.	100, 200, 300	300	Evol. iterations
Max. size	15, 20, 30	30	Max. tree size

subset of daytime operating conditions. Table 2 summarizes the search space and selected values for both methods.

For SVISE, the tuning involves seven parameters that govern different aspects of the model. The Gaussian smoothing width σ controls the degree of pre-filtering applied to the input signal. The polynomial library degree d sets the maximum order of monomial terms available to the model. The global shrinkage parameter τ scales the sparsity-inducing prior on the polynomial coefficients, similar to a Horseshoe prior: smaller values enforce aggressive shrinkage, pushing most coefficients toward zero and producing sparser equations, while larger values permit more terms to remain active. The learning rate η governs the step size of the Adam optimizer during ELBO maximization. The number of radial basis function centers n_τ determines the temporal resolution of the Markov Gaussian process used for state estimation; more centers allow the estimated trajectory to capture higher-frequency variation, while fewer centers enforce smoother paths. The measurement noise level σ_m specifies the assumed observation variance and thereby controls the balance between trusting the raw data and relying on the learned dynamics: a small value forces the estimated state to follow the observations closely, whereas a larger value allows the model to smooth over measurement artifacts. Finally, the number of reparametrization samples n_{rep} controls the Monte Carlo approximation quality when estimating expectations in the ELBO. The full grid comprises 4,050 combinations. Because evaluating a single combination requires training the model on all valid segments in the tuning window, an exhaustive search is computationally prohibitive. We therefore sample 60 unique combinations uniformly from the grid with a fixed random seed for reproducibility.

Training is performed by maximizing the evidence lower bound (ELBO) using the Adam optimizer. We employ the SparsePolynomialIntegratorSDE model variant from the SVISE framework [Course and Nair 2023], which encodes the structural constraint $d\theta/dt = \omega$ directly into the model rather than learning it from data. To improve numerical stability, we apply global standardization to

the state variables and rescale time by a factor of 30. Training uses early stopping with a patience of 300 epochs and a maximum of 10,000 epochs. When the loss diverges to NaN, the model reverts to the last valid checkpoint, resamples the prior weights, and halves the learning rate, allowing up to five such recovery attempts before terminating the run. Gradients are clipped to a maximum norm of 1.0 throughout training. Each hyperparameter combination is evaluated across all valid segments, and the root mean square error (RMSE) of the predicted angular velocity ω serves as the selection criterion. The combination that achieves the lowest mean RMSE across segments is selected and subsequently used to train on all valid five-minute segments across the full dataset.

For PySR, the search space comprises six parameters yielding 3,888 total configurations. The smoothing width σ serves the same denoising role as for SVISE. The tournament size controls how many candidate expressions compete during selection in each evolutionary step; larger tournaments increase selection pressure toward fitter individuals. The parsimony coefficient penalizes expression complexity during evolution, trading accuracy for simpler equations; a value of zero disables this penalty, relying solely on the maximum tree size to constrain complexity. The number of cycles per iteration and the total number of iterations together determine the evolutionary budget, controlling how long the population of expressions is evolved. The maximum expression tree size sets an upper bound on the number of nodes in any candidate expression, limiting structural complexity. PySR is trained separately on each five-minute segment, and configurations are selected based on the trade-off between accuracy and expression complexity, as given by the Pareto front that PySR produces.

5.4 Evaluation Metrics

We first evaluate each method's ability to recover the known governing equations from synthetic data, and then assess predictive accuracy on empirical data using the root mean square error (RMSE) between observed and predicted trajectories. In addition, we examine the physical interpretability of the discovered terms and the complexity of the resulting expressions.

Validation on Synthetic Dataset. On synthetic data, the ground-truth values of the control parameters c_1 and c_2 are known from the Kramers–Moyal estimation used to generate the data, and the constant term corresponds to the power step P_0 of the 1D-L-KM model. Following the validation approach of We verify that the discovered coefficients recover the correct sign and order of magnitude of the known parameters.

Predictive accuracy. The central metric for comparing the three methods is the root mean square error (RMSE) between the observed angular velocity trajectory and the trajectory predicted by the discovered equation within each temporal segment. The RMSE is calculated per segment and then averaged across all valid segments to obtain a single performance measure per method. This metric quantifies how well the discovered deterministic dynamics approximate the measured frequency evolution, independent of the equation's structural form, and therefore provides the most direct

Table 3. Mean RMSE of angular frequency (ω) for our models and benchmark models.

Model	Mean RMSE ω
Our Models	
SVISE	0.095
PySR	0.008436
Benchmark (Wen et al.)	
1D-L-KM	0.292
SINDy Poly(2)	0.142
SINDy Poly(3)	0.153
SINDy Poly(2) + Trig(1)	0.155

basis for comparing the methods' ability to capture the observed dynamics.

Identifying Discovered Dynamics and Equations. On empirical data, we compute the mean absolute value and standard deviation of each coefficient across all temporal segments. These distributions serve as a measure of feature importance and allow a comparison across methods, with the goal of identifying which nonlinear terms consistently emerge with substantial magnitude and may therefore point to complex dynamics in the grid beyond the linear swing equation.

6 Results

6.1 Validation on Synthetic Data

6.2 Application to Empirical Data

7 Discussion

8 Conclusion

References

- M. Anvari, G. Lohmann, M. Wächter, P. Milan, E. Lorenz, D. Heinemann, M. Reza Rahimi Tabar, and Joachim Peinke. 2016. Short term fluctuations of wind and solar power systems. *New Journal of Physics* 18, 6 (June 2016), 063027. doi:10.1088/1367-2630/18/6/063027 arXiv:1606.03426 [physics].
- Steven L. Brunton, Joshua L. Proctor, and J. Nathan Kutz. 2016. Discovering governing equations from data: Sparse identification of nonlinear dynamical systems. *Proceedings of the National Academy of Sciences* 113, 15 (April 2016), 3932–3937. doi:10.1073/pnas.1517384113 arXiv:1509.03580 [math].
- Kevin Course and Prasanth B. Nair. 2023. State estimation of a physical system with unknown governing equations. *Nature* 622, 7982 (Oct. 2023), 261–267. doi:10.1038/s41586-023-06574-8
- Miles Cranmer. 2023. Interpretable Machine Learning for Science with PySR and SymbolicRegression.jl. doi:10.48550/arXiv.2305.01582 arXiv:2305.01582 [astro-ph].
- Sebastian Eichhorn, Anurag Mohapatra, and Christoph Goebel. 2025. PISR: Physics-Informed Symbolic Regression for Predicting Power System Voltage. In *Proceedings of the 16th ACM International Conference on Future and Sustainable Energy Systems*. ACM, Rotterdam Netherlands, 92–107. doi:10.1145/3679240.3734622
- Leonardo Rydin Gorjao, Mehrnaz Anvari, Holger Kantz, Christian Beck, Dirk Witthaut, Marc Timme, and Benjamin Schafer. 2020. Data-Driven Model of the Power-Grid Frequency Dynamics. *IEEE Access* 8 (2020), 43082–43097. doi:10.1109/ACCESS.2020.2967834
- Leonardo Gorjão and Francisco Meirinhos. 2019. kramersmoyal: Kramers–Moyal coefficients for stochastic processes. *Journal of Open Source Software* 4, 44 (Dec. 2019), 1693. doi:10.21105/joss.01693
- Samuel Homan, Niall Mac Dowell, and Solomon Brown. 2021. Grid frequency volatility in future low inertia scenarios: Challenges and mitigation options. *Applied Energy* 290 (May 2021), 116723. doi:10.1016/j.apenergy.2021.116723
- Heqing Huang, Yuzhang Lin, Yifan Zhou, Yue Zhao, Peng Zhang, and Lingling Fan. 2023. Data-driven modeling of power system dynamics: Challenges, state of the art, and future work. *iEnergy* 2, 3 (Sept. 2023), 200–221. doi:10.23919/IEN.2023.0023
- David Kraljic. 2023. Towards Realistic Statistical Models of the Grid Frequency. *IEEE Transactions on Power Systems* 38, 1 (Jan. 2023), 256–266. doi:10.1109/TPWRS.2022.3163336
- Johannes Kruse, Eike Cramer, Benjamin Schäfer, and Dirk Witthaut. 2022. Physics-inspired machine learning for power grid frequency modelling. doi:10.48550/arXiv.2211.01481 arXiv:2211.01481 [eess].
- Ulrich Oberhofer, Leonardo Rydin Gorjao, G. Cigdem Yalcin, Oliver Kamps, Veit Hagenmeyer, and Benjamin Schafer. 2023. Non-linear, bivariate stochastic modelling of power-grid frequency applied to islands. In *2023 IEEE Belgrade PowerTech*. IEEE, Belgrade, Serbia, 1–1. doi:10.1109/PowerTech55446.2023.10202986
- Ulrich Oberhofer, Xinyi Wen, Jaiyong Lee, Heetae Kim, Veit Hagenmeyer, and Benjamin Schäfer. 2025. Nonlinear Stochastic Modeling of the South Korean Power Grid Frequency Dynamics. In *2025 IEEE Kiel PowerTech*. IEEE, Kiel, Germany, 1–6. doi:10.1109/PowerTech59965.2025.11180372
- Leonardo Rydin Gorjão, Richard Jumar, Heiko Maass, Veit Hagenmeyer, G. Cigdem Yalcin, Johannes Kruse, Marc Timme, Christian Beck, Dirk Witthaut, and Benjamin Schäfer. 2020. Open database analysis of scaling and spatio-temporal properties of power grid frequencies. *Nature Communications* 11, 1 (Dec. 2020), 6362. doi:10.1038/s41467-020-19732-7
- Reza Saeed Kandezy, John Jiang, and Di Wu. 2024. On SINDy Approach to Measure-Based Detection of Nonlinear Energy Flows in Power Grids with High Penetration Inverter-Based Renewables. *Energies* 17, 3 (Feb. 2024), 711. doi:10.3390/en17030711
- Andrija T. Saria, Aleksandar A. Saric, Mark K. Transtrum, and Aleksandar M. Stankovic. 2021. Symbolic Regression for Data-Driven Dynamic Model Refinement in Power Systems. *IEEE Transactions on Power Systems* 36, 3 (May 2021), 2390–2402. doi:10.1109/TPWRS.2020.3033261
- Benjamin Schafer, Marc Timme, and Dirk Witthaut. 2018. Isolating the Impact of Trading on Grid Frequency Fluctuations. In *2018 IEEE PES Innovative Smart Grid Technologies Conference Europe (ISGT-Europe)*. 1–5. doi:10.1109/ISGTEurope.2018.8571793
- Iain Staffell and Stefan Pfenninger. 2018. The increasing impact of weather on electricity supply and demand. *Energy* 145 (Feb. 2018), 65–78. doi:10.1016/j.energy.2017.12.051
- Alex M. Stankovic, Aleksandar A. Saric, Andrija T. Saric, and Mark K. Transtrum. 2020. Data-driven Symbolic Regression for Identification of Nonlinear Dynamics in Power Systems. In *2020 IEEE Power & Energy Society General Meeting (PESGM)*. IEEE, Montreal, QC, Canada, 1–5. doi:10.1109/PESGM41954.2020.9281935
- Minh-Quan Tran, Trung Thai Tran, Phuong H. Nguyen, and Guus Pemen. 2023. Sparse identification for model predictive control to support long-term voltage stability. *IET Generation, Transmission & Distribution* 17, 1 (Jan. 2023), 39–51. doi:10.1049/gtd2.12662
- Andreas Ulbig, Theodor S. Borsche, and Göran Andersson. 2015. Analyzing Rotational Inertia, Grid Topology and their Role for Power System Stability. *IFAC-PapersOnLine* 48, 30 (2015), 541–547. doi:10.1016/j.ifacol.2015.12.436
- Xinyi Wen, Mehrnaz Anvari, Leonardo Rydin Gorjao, G. Cigdem Yalcin, Veit Hagenmeyer, and Benjamin Schafer. 2023. Non-standard power grid frequency statistics in Asia, Australia, and Europe. doi:10.48550/arXiv.2308.16842 arXiv:2308.16842 [eess].
- Xinyi Wen, Ulrich Oberhofer, Leonardo Rydin Gorjão, G.Cigdem Yalcin, Veit Hagenmeyer, and Benjamin Schäfer. 2024. Identifying Complex Dynamics of Power Grid Frequency. In *The 15th ACM International Conference on Future and Sustainable Energy Systems*. ACM, Singapore Singapore, 408–414. doi:10.1145/3632775.3661944